Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “voltage” to explain your model or result. Sentence starters are optional.

power: rate of energy transfer

voltage: electric potential difference; energy transferred per charge

current: rate of charge flow

01 Bring back an earlier idea

Parallel branches draw 0.3 A and 0.5 A.
Find the current supplied to their junction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: P = IV; E = PΔt; P = I²R = V²/R.
Constant power for E = PΔt; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power VI=12 W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Two motors use the same energy but one takes half the time.
Compare average power.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Build the solution

A motor draws 2 A at 6 V.
Find P=2×6, then energy used in 5 s using E=P×5.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A wave travels at fixed speed. A second target is 4 times farther away than the first.
Compare one-way travel times.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

A 3 W device runs for 10 s.
Find energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two devices use the same 60 J: one runs at 10 W, another at 20 W.
Find their run times.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3